

Hallucination Failure Types

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Multimodal Evidence

Without opening the old lesson, explain one key idea from Multimodal Evidence. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Visual Composition

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Structured Output and Schemas

Without opening the old lesson, explain one key idea from Structured Output and Schemas. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Generation failures and verification. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Generation failures and verification. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong input. Classify the three failures and their repairs.

6. Solve the given case

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

Separate the failures and checks.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

An answer gives a book title that does not exist and quotes words absent from the cited passage. Identify two failures and a repair.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 14 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong input. Classify the three failures and their repairs.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: Source agreement and arithmetic validity are separate checks; missing evidence is not proof of the opposite claim. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Hallucination Failure Types

A fictional Study Buddy drafts a field-trip summary from one approved record.

Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

Hallucination Failure Types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute duration

12 / 4 = 3 hours, excluding stops.



2. Contradicted claim

Two-hour duration disagrees with the calculation from S.



3. Supported claim

Departure 09:00 is stated in S.



4. Unsupported claim

Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

CHANGE ONE THING

Add a planned 30-minute stop while preserving speed and distance.

Explain the changed result using the matching lesson.

Can we conclude there will definitely be no lunch from S alone?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Audit Study Buddy outputs claim by claim, recording contradiction, missing support and correct calculations separately.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Audit Study Buddy outputs claim by claim, recording contradiction, missing support and correct calculations separately.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy drafts a field-trip summary from one approved record.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

1. Unsupported claim
2. Supported claim
3. Contradicted claim
4. Compute duration

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Unsupported claim Free lunch has no source; missing evidence is not a confirmed no-lunch fact. Different errors need different repairs. Recalculate the duration; keep the supported departure; remove or seek evidence for lunch. These categories are more informative than calling the whole answer wrong. Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Practice 2: explain the deciding evidence

Changed-case practice: Add a planned 30-minute stop while preserving speed and distance. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Planned total elapsed time becomes 3.5 hours.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Planned total elapsed time becomes 3.5 hours. Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Repair a missing building block

Review Multimodal Evidence (ai-3/13). Goal: Explain the mechanism and terminology of speech pipelines and multimodal evidence. Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Explain the mechanism and terminology of image generation and provenance. Short practice: Which checks pass after that single change, and what still needs repair?. Return to this week and explain what you changed.

Review Structured Output and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
